

First Steps with a Full Node

made by Dongguri-Jun

contact : hello@dongguri.org

Licence : CC BY-SA 4.0

v1.0

2026.08.16 last edited

Map

Full Node Fundamentals

1. What is Bitcoin?
2. What is a full node?
3. The meaning of running a full node?

Why Run a Full Node?

1. Why the motivation feels unclear (the rewards moved elsewhere)
2. Common reasons
3. Privacy
4. Sovereignty
5. Network contribution
6. The overlooked reason
7. Learning through hands-on practice

Getting Started

1. Recommended hardware
2. DIY vs prebuilt
3. Choose and install an implementation
4. Synchronization

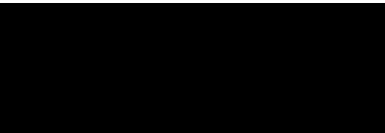
Using Your Full Node

1. Connect a watch-only wallet
2. Join the network
3. Lightning
4. Mining

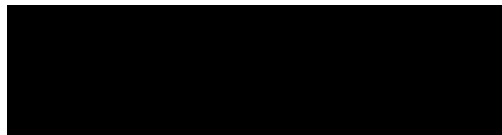


Intro

If you don't run a full node, are you really a Bitcoiner...?

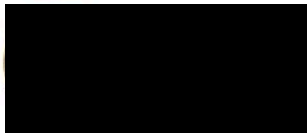


[번역 보기](#)
If you don't run a node, do you even Bitcoin?



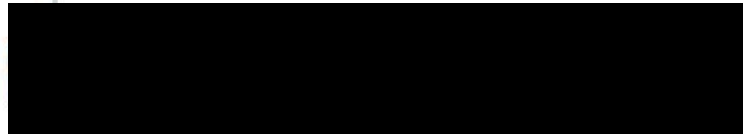
[번역 보기](#)
If you don't run a node, you're not a bitcoiner.
You couldn't wait to build your own PC in the 90's.
Run a goddam node and connect your bitcoin wallet.

[@Raspberry_Pi](#) 4 [@pimoroni](#) NVME HAT and [@umbrel](#) OS.



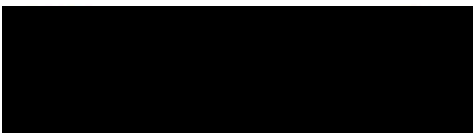
[번역 보기](#)
A bitcoiner without a node is not a bitcoiner.

오전 12:18 · 2026년 2월 27일 · 2,311 조회수



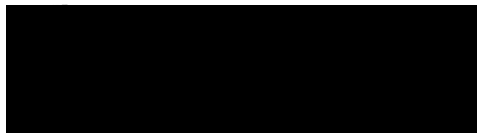
[번역 보기](#)
I'll be honest with you, and not attacking you personally so don't take it the wrong way-- anyone who doesn't run a Bitcoin node is not a real Bitcoiner. If you don't run a node, it's impossible to even know if you own real BTC, without having to take someone else's word for it

오후 10:34 · 2026년 6월 30일 · 122 조회수



[번역 보기](#)
If you're not running a node, mining, or in self custody you're not a bitcoiner. So you might wanna take BTC out of your Twitter name

오후 1:17 · 2026년 5월 26일 · 162 조회수



[번역 보기](#)
And: SET UP A NODE!
(Ideally the [@start9labs](#))

No own node = not a real Bitcoiner (just a tourist)

오후 5:48 · 2026년 7월 23일 · 75 조회수



Full Node Fundamentals — What Is Bitcoin?

Can money move without intermediaries?

Traditional finance — intermediary layers

Central bank · commercial banks · clearinghouses · SWIFT · card networks

\$2.5T per year

Intermediation costs paid to move money (3.6T transactions)

6.4% · Banks 9.5%

Average fee on \$200 remittance — 2–3× UN target (3%)

Bitcoin — only the protocol

Rules + participants (wallets · miners · full nodes)

P2P transfer

Within the protocol, value can move P2P without intermediaries

Transaction size

Fees usually depend on transaction size, not the amount sent.

Who performs all those institutional functions in Bitcoin?



Full Node Fundamentals — What Is Bitcoin?

Traditional finance vs Bitcoin

An explanatory comparison; the systems differ in important ways.

Traditional finance — roles by institution

Central bank · government

Set issuance and monetary policy

Central bank

Issue currency

Commercial banks

Accounts · transfers · custody of balances

Banks · clearinghouses

Ledger records · transaction validation · settlement



Bitcoin — roles of rules and participants

Protocol rules

21 million cap, fixed in code

Miners

Issue new coins only within protocol rules

Wallet

Manage keys · create/sign transactions (self-custody)

Full node

Check transaction/block rules · keep a chain copy · relay

Question: “Does a miner’s block enter the ledger immediately?”

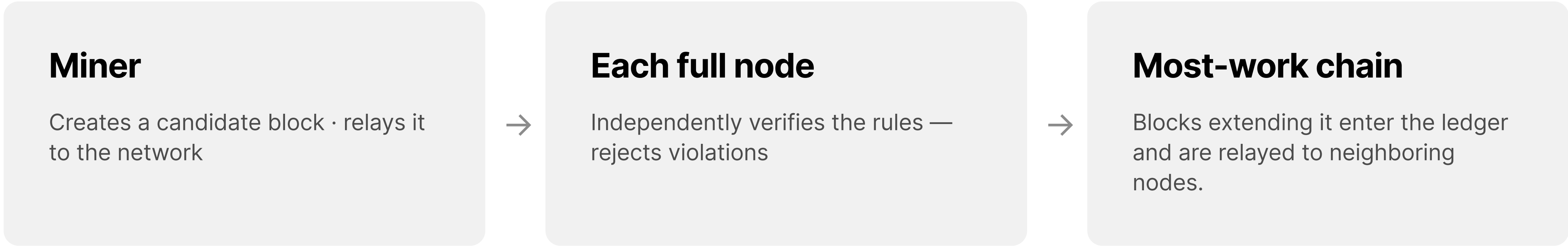


Full Node Fundamentals — What Is Bitcoin?

Does a miner's block equal a final ledger entry?

Each full node verifies the rules.

Only blocks on the chain with the greatest cumulative work are recorded.



A system where every participant verifies for themselves



Full Node Fundamentals — What Is a Full Node?

What a full node does — validate, store, relay

Validation

Check transactions and blocks
the node receives against the rules

Storage

Keep the chain data needed
for validation on your device

Relay

Pass only valid data
to neighboring nodes

Now, each of these three roles in detail



Full Node Fundamentals — What Is a Full Node?

① Validation — Four checks

Scope: transactions and blocks received from the network — verify again even if others did

Did the owner authorize this spend?

(Signature)

```
// script/interpreter.cpp
checker.CheckECDSASignature(...) // ECDSA example; Schnorr too
```

Is the same money being spent twice?

(Double-spend)

```
// consensus/tx_verify.cpp
if (coin.IsSpent()) return state.Invalid(...);
```

Was this block produced by real mining?

(Proof of work)

```
// pow.cpp — CheckProofOfWork
if (UIntToArith256(hash) > bnTarget)
    return false;
```

Does it obey the 21 million cap?

(Consensus rules)

```
// validation.cpp — ConnectBlock (MAX_MONEY = amount range check)
if (block.vtx[0]→GetValueOut() > blockReward)
    state.Invalid(..., "bad-cb-amount");
```

Any violation = immediate rejection · only valid data moves to storage →



Full Node Fundamentals — What Is a Full Node?

② Storage — verify the full history on your device

What

Initially verifies every block
since the first block in 2009
(about 750 GB for a full node)

How

Unpruned full node = keeps full
history
Pruned full node = discards old
block data

Why

Verify directly from my own ledger
without asking anyone else



Full Node Fundamentals — What Is a Full Node?

③ Relay — pass only valid data to neighbors



Only transactions and blocks that pass my validation go to neighbors — each neighbor verifies again

Validating nodes reject rule violations · valid transactions and blocks usually propagate quickly



Full Node Fundamentals — Running a Full Node

Running a full node

Set up a node on your own device to perform the three roles above (store, validate, relay) and join the network

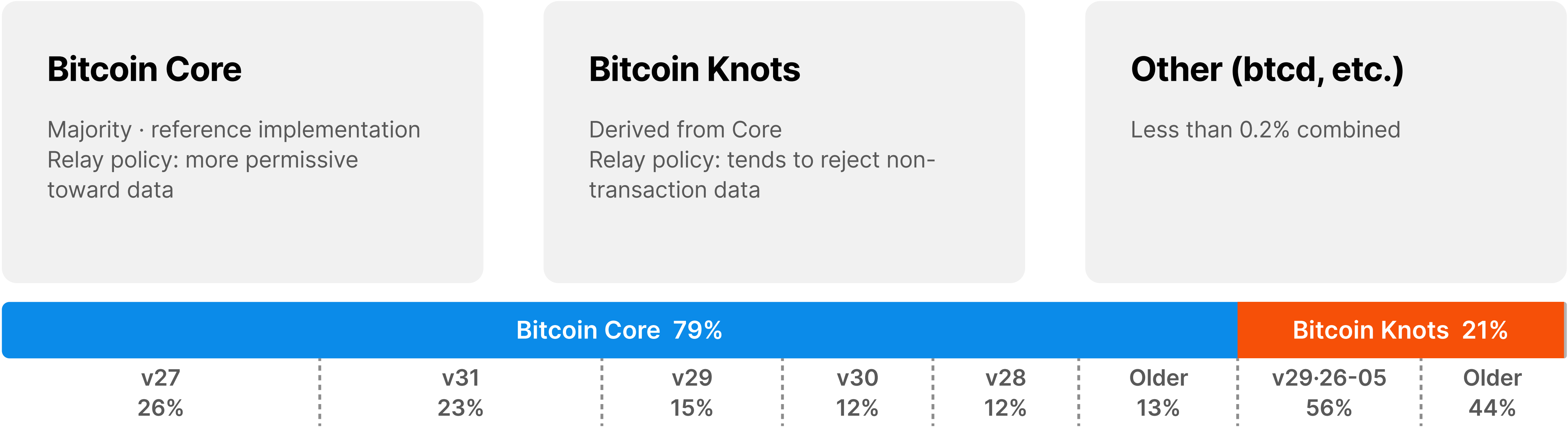
Method = install Bitcoin software on your device

No qualifications or permission — installation makes you part of the network



Full Node Fundamentals — Running a Full Node

Implementation — software that encodes the rules



Why Run a Full Node? — Is It Worth It?

Do I really need to run a full node?

Mining → earns rewards

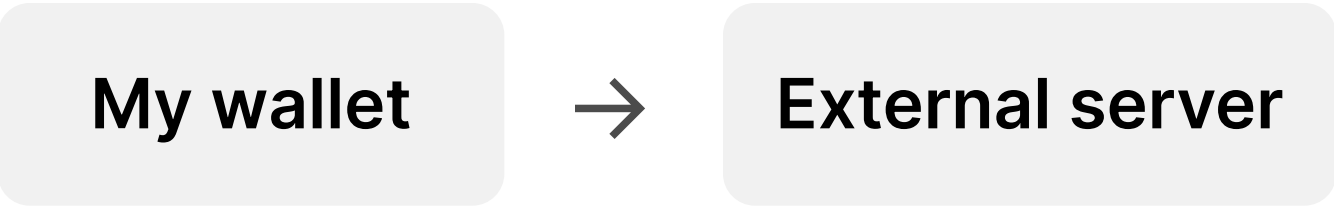
Wallet → essential for using Bitcoin

Full node → no obvious immediate problem without one

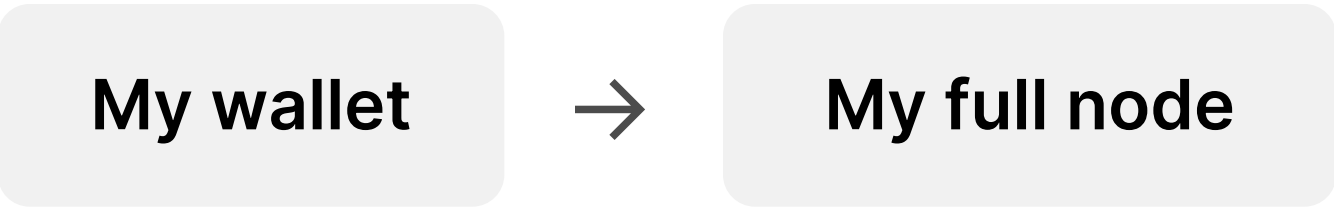


Why Run a Full Node? — Common Reasons

① Privacy



Balances · transactions · IP exposed = surveillance



Less exposure from external queries

External server

Addresses, transaction history, and connection data may be collected in one place. Combined with other records, they may help link wallet activity. The server may omit results or refuse to broadcast a transaction.

My own server

Less exposure from asking an external server about addresses and transactions.

Remaining limits

On-chain records stay public. Protecting your IP requires Tor or other settings.

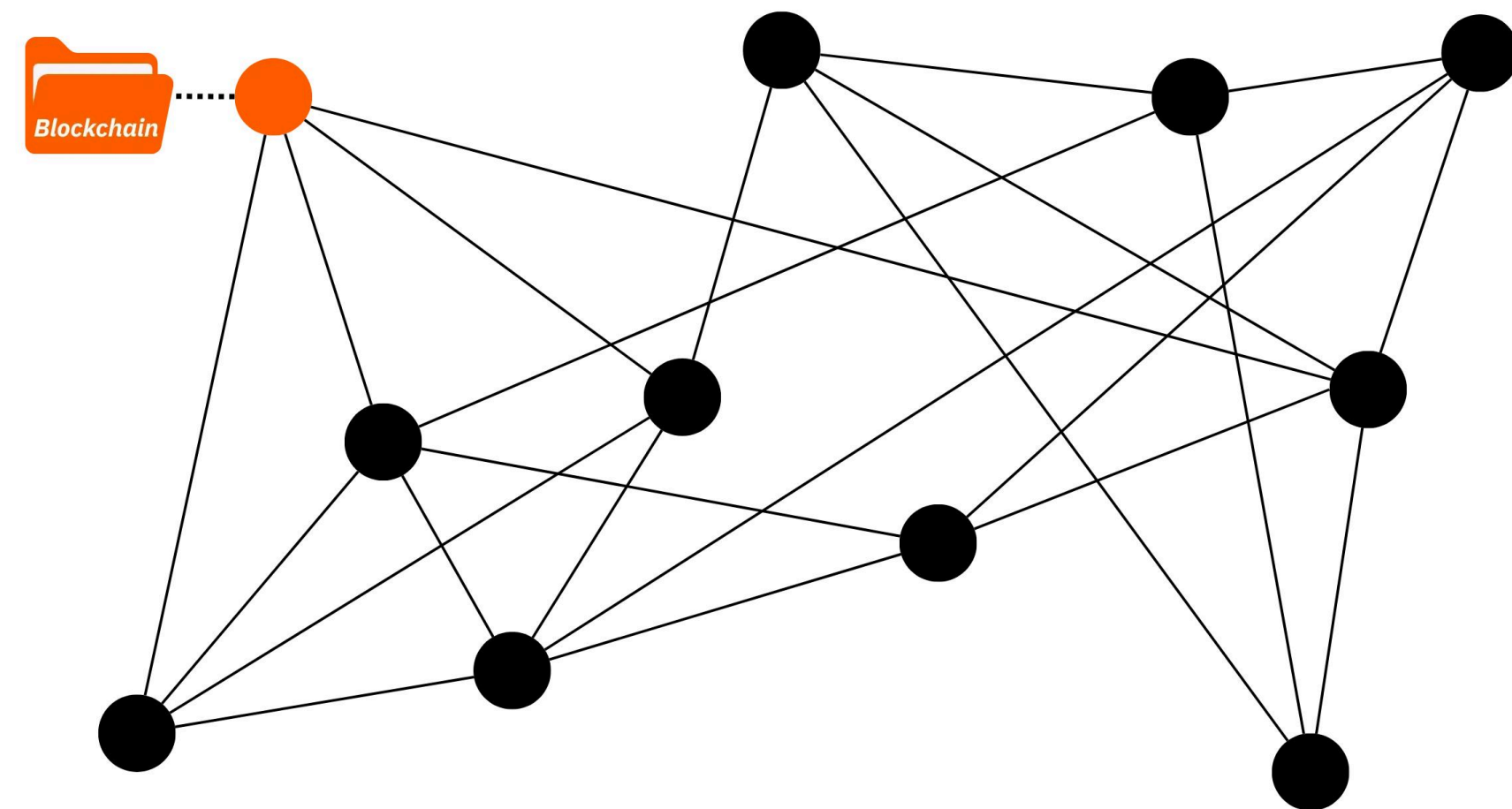
Source: Electrum FAQ · ElectrumX Protocol · 2026.07.30

Image: Plan B Network · CC BY-SA 4.0



Why Run a Full Node? — Common Reasons

② Verification sovereignty



Value

Verify directly without trusting others — “Don’t trust, verify” in practice

In practice

Check balances and transactions against the chain verified by my full node. Reject transactions and blocks that break the rules (“Verify it yourself”)

Why it feels distant

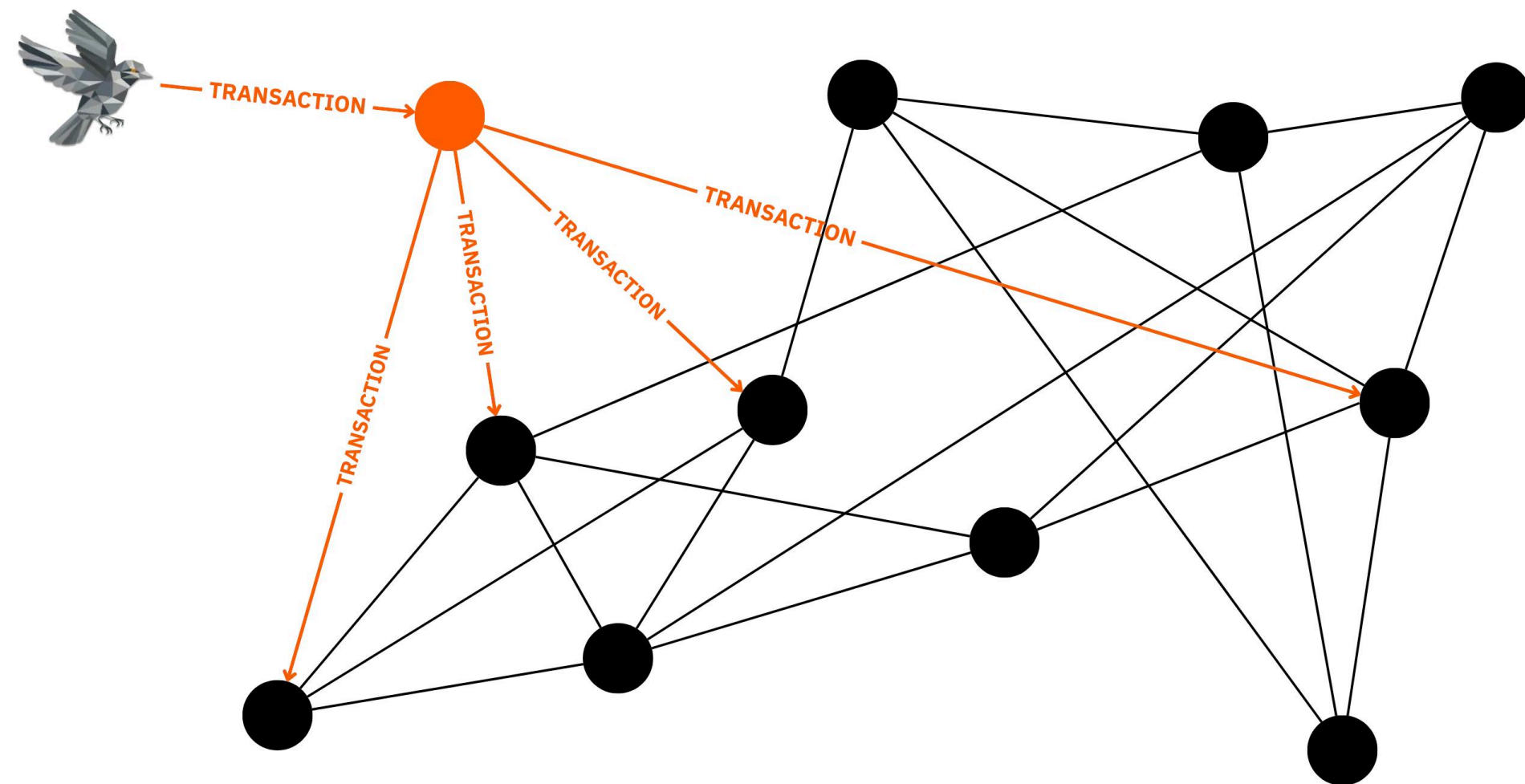
Personal sovereignty feels abstract

It is hard to tell clearly whether you have reclaimed it



Why Run a Full Node? — Common Reasons

③ Network contribution



Value

One more node = stronger decentralization and resilience

In practice

A publicly reachable node can help new nodes synchronize and add capacity for relaying transactions and blocks.

More redundancy and resilience

Why it feels distant

An action for everyone, with no direct personal return

The network is already robust



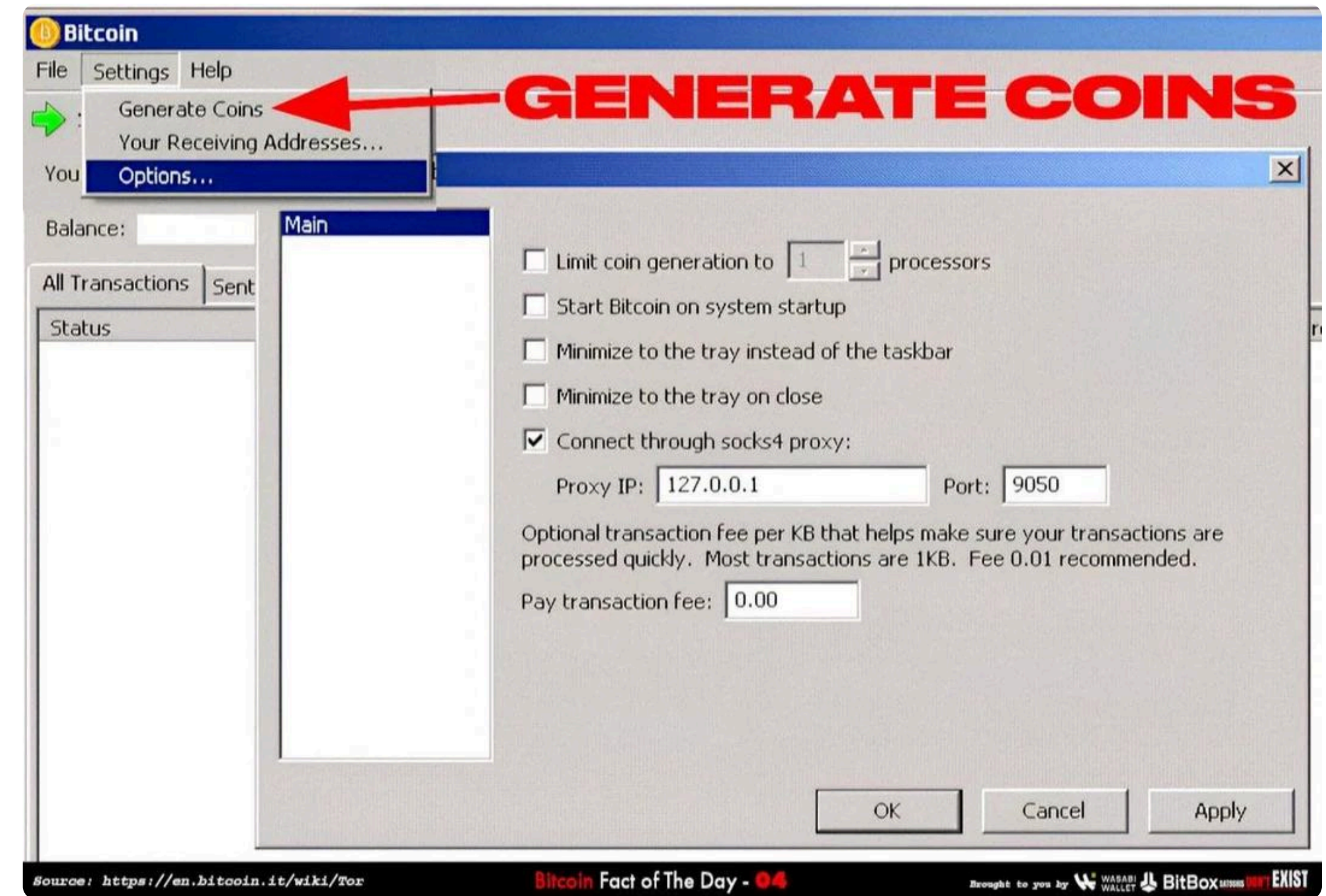
Why Run a Full Node? — Bitcoin's Early History

Originally, all in one

Early Bitcoin = wallet + mining + full node in one program

Leave it running to validate and earn mining rewards

→ a clear economic incentive



Why Run a Full Node? — Mining and Wallets Specialize

Mining

Industrialized through ASICs and large mining farms
(the rewards moved here)

Wallet

Light and mobile wallets
(the convenience moved here)

Full node

No separate reward
(only the validator role remains)

Mining = industrialized (cheap power + ASIC economies of scale)

Three ASIC makers (Bitmain · MicroBT · Canaan) > 99%

Public miners account for about 47% of network hash rate (top five ≈ 29%) — the rest are private or small-scale

Regional concentration: North America (75% U.S. among firms responding to the Cambridge survey)

The incentive-bearing functions split apart



Why Run a Full Node? — The Overlooked Reason

An asset with no third party responsible — learn it yourself

Banks and brokers carry institutional responsibility · decentralized Bitcoin places responsibility on me

Ignorance creates anxiety

Hacking · quantum computers · crashes → people who study can answer for themselves

Affordable self-investment

Store your life savings for 10 or 20 years in something you barely know?
Study is basic investing — a few hundred thousand KRW (node · books · classes) · lower time preference

**Bitcoin is a tool
for storing the value of your hard-earned labor**



Why Run a Full Node? — The Overlooked Reason

A full node = hands-on practice for learning Bitcoin deeply

Turn reading into hands-on understanding

Touch the core

Validate · store · relay yourself —
experience Bitcoin's heartbeat

See the whole process

Installation · IBD · validation · logs ·
terminal — broader protocol practice
than a book alone

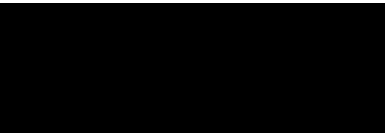
Practical effect

Greater confidence in tools · less
exposure to errors and scams

A full node = hands-on practice “for me”

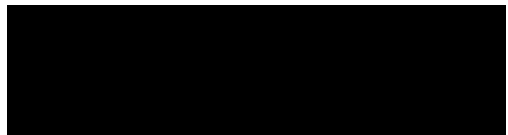


The real meaning Running a full node as you study helps you understand Bitcoin properly.



번역 보기

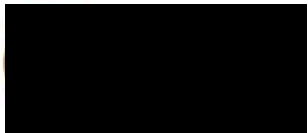
If you don't run a node, do you even Bitcoin?



번역 보기

If you don't run a node, you're not a bitcoiner.
You couldn't wait to build your own PC in the 90's.
Run a goddam node and connect your bitcoin wallet.

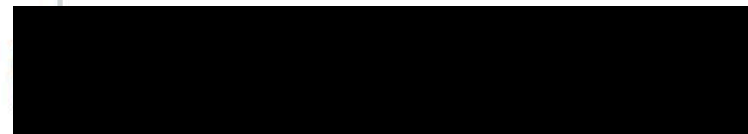
@Raspberry_Pi 4 @pimoroni NVME HAT and @umbrel OS.



번역 보기

A bitcoiner without a node is not a bitcoiner.

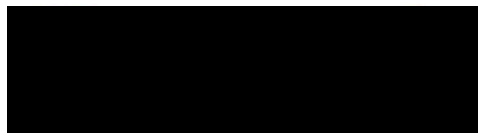
오전 12:18 · 2026년 2월 27일 · 2,311 조회수



번역 보기

I'll be honest with you, and not attacking you personally so don't take it the wrong way-- anyone who doesn't run a Bitcoin node is not a real Bitcoiner. If you don't run a node, it's impossible to even know if you own real BTC, without having to take someone else's word for it

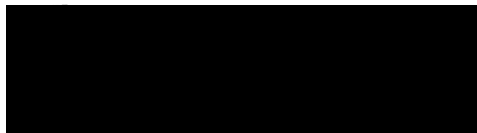
오후 10:34 · 2026년 6월 30일 · 122 조회수



번역 보기

If you're not running a node, mining, or in self custody you're not a bitcoiner. So you might wanna take BTC out of your Twitter name

오후 1:17 · 2026년 5월 26일 · 162 조회수



번역 보기

And: SET UP A NODE!
(Ideally the @start9labs)

No own node = not a real Bitcoiner (just a tourist)

오후 5:48 · 2026년 7월 23일 · 75 조회수



Full Node Journey — Practice Roadmap

Full Node Practice Roadmap

From choosing hardware to connecting a wallet and exploring further



Getting Started — STEP 1 Prepare Hardware

Recommended full-node setup — no high-end computer needed

CPU

64-bit dual-core or better
(quad-core recommended)

RAM

4 GB or more
(8 GB recommended)

Storage

2 TB or more
SSD

Start with a mini PC or an old laptop + 2 TB SSD



Getting Started — STEP 1 Prepare Hardware

DIY vs prebuilt

Option 1) Build it yourself

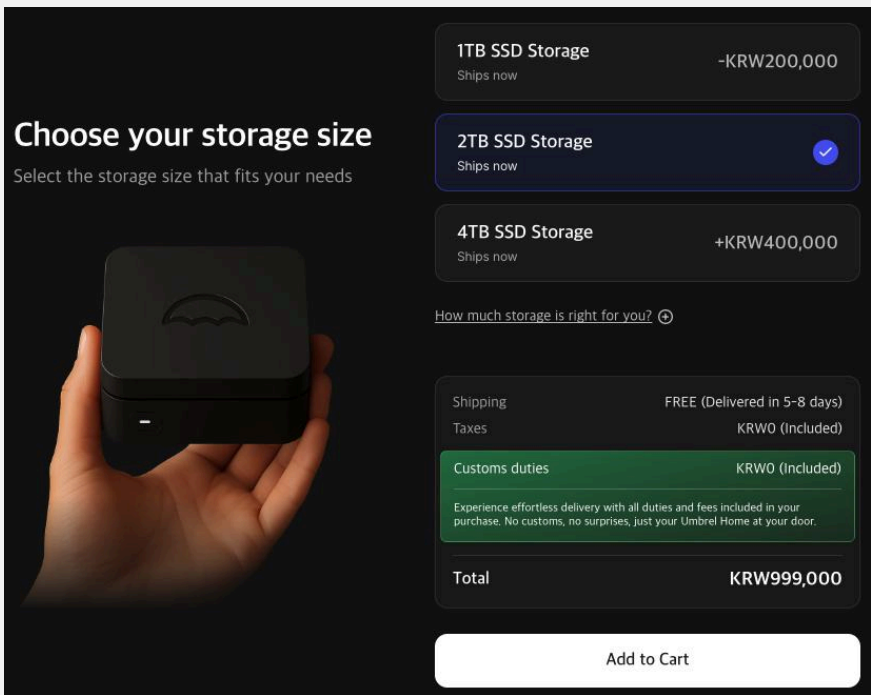


**2 TB NVMe
SSD**

**Used
computer**

Current budget build: KRW 630,000
Reuse spare parts to lower the cost further

Option 2) Prebuilt (recommended)



Umbrel Home currently KRW 999,000
Minimal setup burden; recommended for most people

Hardware revenue funds software maintenance → an investment in sustainability

Source: Compuzone · Umbrel · Korean prices / official product ·

2026.07.30



Getting Started — STEP 2 Install · Synchronize

Four installation steps Example: Umbrel OS

Step 1

Install Umbrel OS on the device (prebuilt units arrive with it installed)

Step 2

Connect through a browser on the same network (umbrel.local)

Step 3

Download apps from the App Store

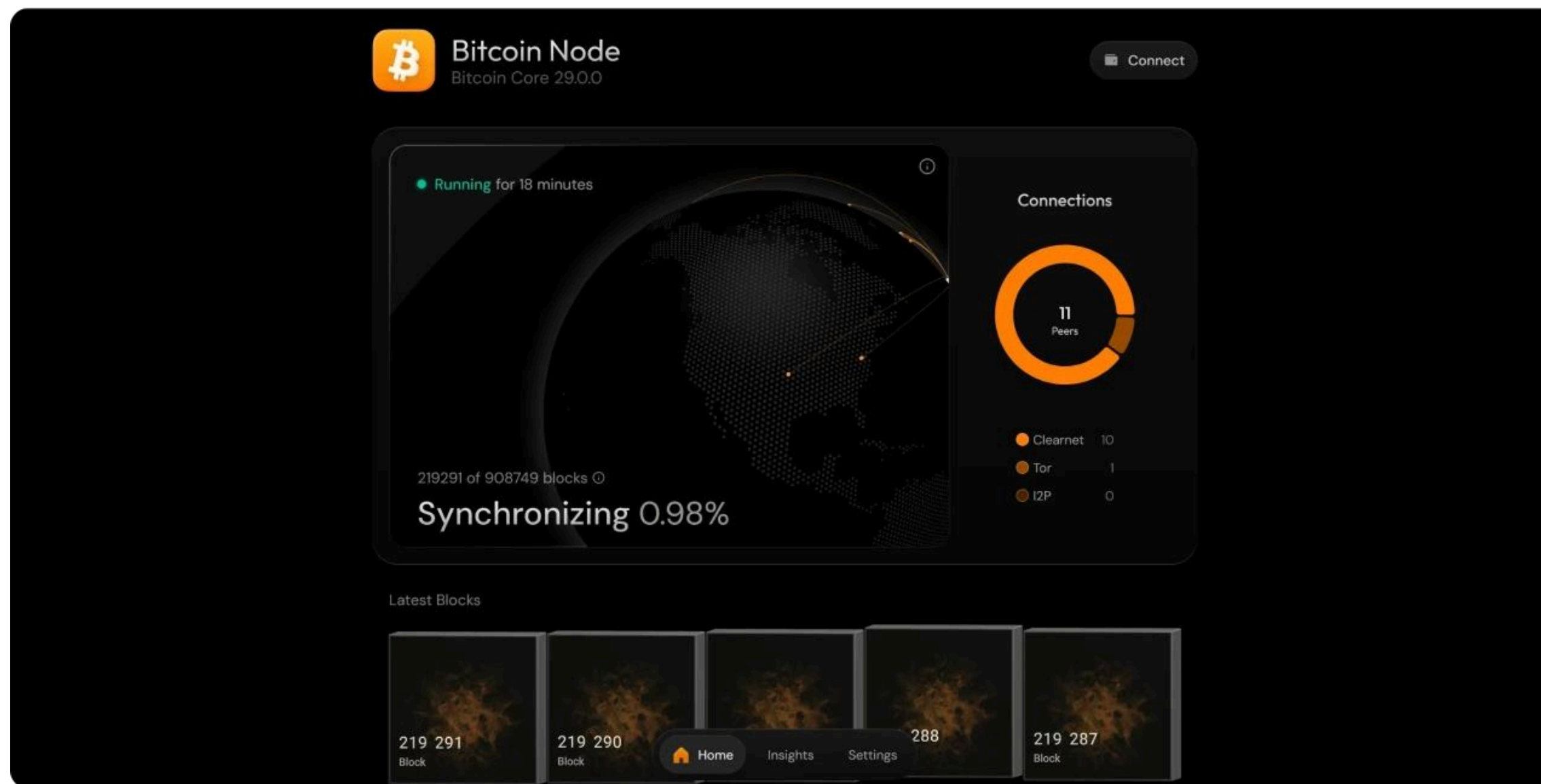
Step 4

Use IBD to download and validate the ledger



Getting Started — STEP 2 Install · Synchronize

Synchronization (IBD) Initial Block Download



Receive the full ledger from the first block in 2009 (the genesis block)

More than a download — every block is validated as it arrives

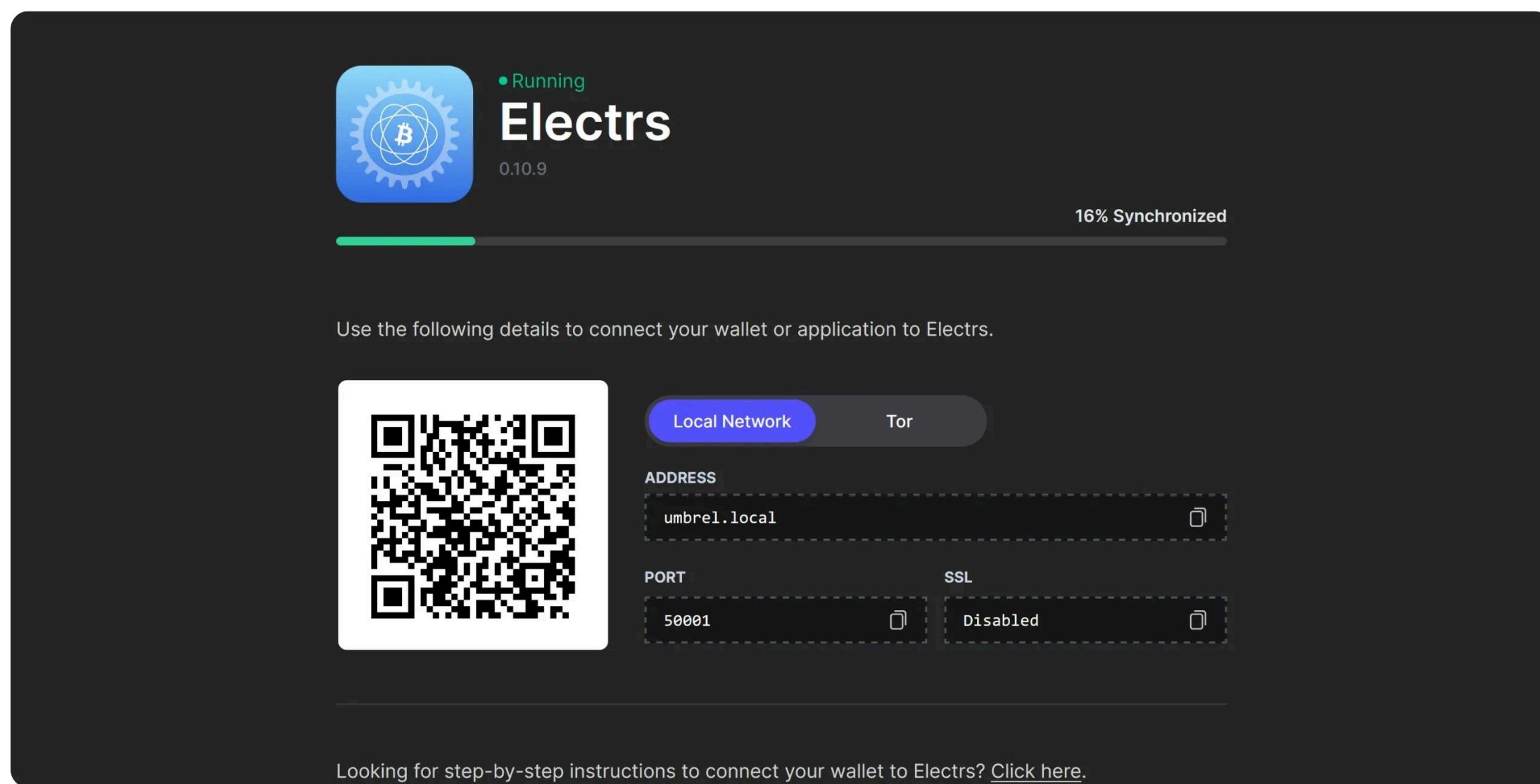
Usually one day to one week, depending on computer performance

Synchronization complete = a full node ready to contribute to the network



Using Your Full Node — STEP 3 Connect a Wallet

Connect your wallet to put your node to practical use



Install Electrs (an Electrum server)

A helper app that lets your wallet quickly search ledger data on your node

Tailscale (private VPN) or Tor

Connect securely to your node from outside your home network

Connect a watch-only wallet

Enter your Electrs endpoint in Coconut Wallet or BlueWallet



Using Your Full Node — STEP 3 Connect a Wallet

Sovereignty over your wallet's query path

Before

My wallet → queries someone else's server
(address and transaction history may be exposed)

After

My wallet → queries my node
(less exposure from external address and transaction lookups)

Less exposure to external server queries · on-chain visibility and IP privacy need separate settings



Using Your Full Node — STEP 4 Join the Network

**Joining the network
= my choice**

Allow inbound connections

Accept connections from other nodes. Publicly reachable nodes can help new nodes sync and relay transactions and blocks.

Core port 8333 · check NAT, router, and firewall settings.

Implementation/version choice = my signal

A personal choice of transaction relay policy

Separate from consensus-rule changes or a formal vote

Choose the policies my node will validate and relay



Using Your Full Node — STEP 5 Explore Further

Solo mining



Bitaxe 602 Gamma

Bitaxe → my Stratum server → my full node → Bitcoin network

Stratum server (connects the mining device to the node)

Receive block templates and mine

If you find one, check reward and fee terms



Three takeaways

1

**Bitcoin lets anyone
manage and verify
for themselves**

2

**A full node is a leading
hands-on way to learn
Bitcoin deeply.**

**Reduce uncertainty and
explore privacy and
sovereignty
firsthand.**

3

**Start at whichever stage
interests you — no
high-end
hardware or coding
required**

